

Domain and Range from Problem Situations

TEACHER KEY | Algebra II | Unit 1 | Day 2 | TEKS A2.7.I, A2.2.A

Objective: I can find the domain and range of a real-world situation, explain what they mean in context, and tell the difference between the domain of a function and the domain that the situation allows.

Part 1 — Name the Variables

Which quantity is the input and which is the output?

- Hours you work and the money you earn
independent: **hours worked** dependent: **money earned**
- Number of tickets bought and the total cost
independent: **tickets bought** dependent: **total cost**
- Seconds since a ball was thrown and its height
independent: **seconds** dependent: **height**

Part 2 — Function Domain vs. Contextual Domain

A cube-shaped box has side length x inches. Its volume is $V(x) = x^3$.

What domain does the **equation** x^3 accept? **all real numbers**

What domain does the **box** allow? **$x > 0$, or $(0, \infty)$**

Why? **a side length cannot be zero or negative**

Big idea: the equation may accept numbers the **situation** does not. Always ask what makes sense.

Part 3 — Domain and Range in Context

Give the contextual domain and range. Use an inequality and interval notation.

Example 1. Concert tickets cost \$12 each. You may buy at most 8. Total cost $C(t) = 12t$.

Domain: **$t = 0, 1, 2, \dots, 8$ (whole tickets only)** Discrete or continuous? **discrete**

Range: **$\{0, 12, 24, 36, 48, 60, 72, 84, 96\}$**

You Try 1. A ball is thrown. Its height is $h(t) = -16t^2 + 64t$ feet, and it is in the air from $t = 0$ to $t = 4$ seconds. Its greatest height is 64 feet.

Domain: **$0 \leq t \leq 4$** interval **$[0, 4]$** Discrete or continuous? **continuous**

Range: **$0 \leq h \leq 64$** interval **$[0, 64]$**

You Try 2. A parking garage charges \$3 per hour and is open 6 a.m. to 10 p.m. (16 hours).

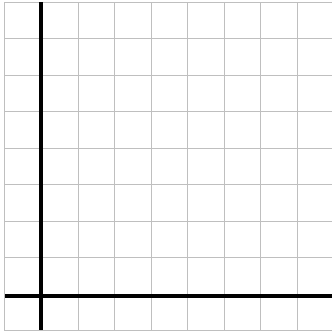
Domain: **$0 \leq h \leq 16$, $[0, 16]$** Range: **$0 \leq c \leq 48$, $[0, 48]$**

Part 4 — Reading a Situation from a Graph

Each square is 1 unit. Plot the points, then answer in context.

A water tank is drained. Gallons g after m minutes:

m	0	1	2	3	4
g	8	6	4	2	0



Answer in context.

Domain: $0 \leq m \leq 4$, $[0, 4]$

Range: $0 \leq g \leq 8$, $[0, 8]$

What does the domain mean here?

the tank drains over 4 minutes

What does a range of 0 mean?

the tank is empty

Part 5 — Does the Situation Restrict the Domain?

Write yes or no, and if yes, give the restriction.

- The side length of a square garden **yes, $x > 0$**
- The temperature in a freezer over one day **no, it may be negative**
- The number of buses needed for a field trip **yes, whole numbers only**

Practice

Give the contextual domain and range for each. Show your thinking on the lines.

- A phone plan costs \$30 per month plus \$5 per gigabyte, up to 20 GB.

Domain $0 \leq g \leq 20$, $[0, 20]$ Range $30 \leq c \leq 130$, $[30, 130]$

- A theater has 250 seats. Tickets are \$9 each, and t tickets are sold.

Domain $t = 0, 1, 2, \dots, 250$ Range $\{0, 9, 18, \dots, 2250\}$

- A diver descends from the surface to 40 meters below sea level.

Range of depth $-40 \leq d \leq 0$, $[-40, 0]$ Why negative? **below sea level**

Exit Ticket

A taxi charges a \$4 flat fee plus \$2 per mile, for trips up to 10 miles. Cost $C(m) = 4 + 2m$.

Domain $0 \leq m \leq 10$, $[0, 10]$ Range $4 \leq C \leq 24$, $[4, 24]$

Why does the range start at 4, not 0? **the \$4 flat fee is charged even at 0 miles**